

Predictive Maintenance for Electric Vehicle Charging Station

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Abstract

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1 Problem description

1.1 Optimization model

1.2 Problem description

We consider a fleet maintenance planning problem over a finite planning horizon. Let I denote the set of fleet assets, F the set of failure types, and D the set of discrete time periods (days) within the planning horizon. For each asset $i \in I$ and failure type $f \in F$, a prognostic model provides a predicted remaining useful life RUL_{if} , interpreted as the predicted time-to-failure (in days) if no intervention is performed.

The decision problem consists of selecting, for each pair (i, f) , a maintenance execution day $d \in D$. Initiating a maintenance action on asset i incurs a setup cost c_i^{setup} . Depending on the selected timing, maintenance is classified into one of three modes: *regular predictive maintenance*, *emergency predictive maintenance*, or *reactive maintenance*, with costs c_{if}^{pr} , c_{if}^{u-pr} , and c_{if}^{re} , respectively. In addition, the model accounts for timing deviations through nonnegative penalty variables: φ_{if} (tardiness / downtime) weighted by α_{if} and ψ_{if} (earliness / unused RUL) weighted by β_{if} . The objective is to minimize the total cost over the planning horizon, combining setup costs, mode-dependent costs, and timing-penalty costs.

To restrict attention to failure events within the planning horizon, we define a binary parameter F_{if} indicating whether failure type f on asset i is expected to occur during the horizon (i.e., $F_{if} = 1$ if $RUL_{if} \leq |D|$, and $F_{if} = 0$ otherwise). The complete notation is summarized below.

Sets

- I The set of fleet assets (index i).
- F The set of failure types (index f).
- D The set of days in the planning horizon (index d).

Parameters

Maintenance

- c_i^{setup} The setup cost for initiating maintenance on asset i .
- c_{if}^{pr} The cost of *regular predictive maintenance* for asset i and failure type f .
- c_{if}^{u-pr} The cost of *emergency predictive maintenance* for asset i and failure type f .
- c_{if}^{re} The cost of *reactive maintenance* for asset i and failure type f .
- α_{if} Penalty weight for tardiness / downtime variable φ_{if} for (i, f) .
- β_{if} Penalty weight for earliness / unused-RUL variable ψ_{if} for (i, f) .
- RUL_{if} Predicted remaining useful life (days) of asset i for failure type f .
- F_{if} Binary indicator equal to 1 if failure type f on asset i occurs within the planning horizon, i.e., $RUL_{if} \leq |D|$.
- LT_i Lead time (days) required to plan and initiate maintenance on asset i .
- R Maximum number of assets that can be initiated per day (daily capacity).
- M A sufficiently large constant used in big- M logical constraints.

Decision variables

x_{ifd}	1 if maintenance for asset i and failure type f is scheduled on day d , and 0 otherwise.
y_{id}	1 if maintenance of asset i is initiated on day d , and 0 otherwise.
e_{if}	1 if the plan selects <i>regular predictive maintenance</i> for (i, f) , and 0 otherwise.
l_{if}	1 if the plan selects <i>reactive maintenance</i> for (i, f) , and 0 otherwise.
u_{if}	1 if the plan selects <i>emergency predictive maintenance</i> for (i, f) , and 0 otherwise.
φ_{if}	Nonnegative tardiness / downtime amount for (i, f) .
ψ_{if}	Nonnegative earliness / unused-RUL amount for (i, f) .

Thus the objective of the problem is as follows:

$$\begin{aligned} \text{Min } Z = & \sum_{i \in I} \sum_{d \in D} c_i^{\text{setup}} y_{id} + \sum_{i \in I} \sum_{f \in F} \alpha_{if} \varphi_{if} + \sum_{i \in I} \sum_{f \in F} \beta_{if} \psi_{if} \\ & + \sum_{i \in I} \sum_{f \in F} c_{if}^{\text{pr}} e_{if} + \sum_{i \in I} \sum_{f \in F} c_{if}^{\text{u-pr}} u_{if} + \sum_{i \in I} \sum_{f \in F} c_{if}^{\text{re}} l_{if}. \end{aligned} \quad (1)$$

Objective (1) minimizes the total maintenance planning cost. It includes the setup cost c_i^{setup} whenever maintenance on asset i is initiated on day d (via y_{id}), adds timing-penalty costs $\alpha_{if}\varphi_{if}$ and $\beta_{if}\psi_{if}$, and includes the mode-dependent maintenance costs: c_{if}^{pr} (via e_{if}), $c_{if}^{\text{u-pr}}$ (via u_{if}), and c_{if}^{re} (via l_{if}).

$$\sum_{d \in D} x_{ifd} = F_{if} \quad \forall i \in I, f \in F \quad (2)$$

$$\sum_{f \in F} x_{ifd} \leq M y_{id} \quad \forall i \in I, d \in D \quad (3)$$

$$\sum_{i \in I} y_{id} \leq R \quad \forall d \in D \quad (4)$$

$$\sum_{d \in D} d x_{ifd} - RUL_{if} \leq \varphi_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (5)$$

$$(RUL_{if} - LT_i) - \sum_{d \in D} d x_{ifd} \leq \psi_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (6)$$

$$\varphi_{if} \leq M l_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (7)$$

$$\psi_{if} \leq M e_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (8)$$

$$l_{if} + e_{if} + u_{if} = 1 \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (9)$$

$$x_{ifd}, l_{if}, e_{if}, u_{if} \in \{0, 1\} \quad \forall i \in I, f \in F, d \in D \quad (10)$$

$$\varphi_{if} \geq 0, \psi_{if} \geq 0 \quad \forall i \in I, f \in F \quad (11)$$

Constraint (2) enforces the maintenance scheduling requirement for each asset–failure pair (i, f) . If $F_{if} = 1$, exactly one day d must be selected such that $x_{ifd} = 1$. If $F_{if} = 0$, no maintenance is scheduled within the horizon and $x_{ifd} = 0$ for all $d \in D$.

Constraint (7) activates reactive maintenance through a big- M formulation: positive tardiness (captured by φ_{if}) is only allowed when $l_{if} = 1$, which selects reactive maintenance and triggers cost c_{if}^{re} .

Constraint (8) activates regular predictive maintenance through a big- M formulation: positive earliness (captured by ψ_{if}) is only allowed when $e_{if} = 1$, which classifies the decision as regular predictive maintenance and triggers cost c_{if}^{pr} .

Constraint (9) ensures exclusive mode selection for each (i, f) by forcing exactly one of the three indicators to be active: regular predictive ($e_{if} = 1$), reactive ($l_{if} = 1$), or emergency predictive ($u_{if} = 1$).

Constraints (5)–(6) define the nonnegative timing-penalty variables φ_{if} and ψ_{if} as upper bounds on lateness and earliness (relative to the corresponding timing references), which are weighted in the objective by α_{if} and β_{if} , respectively. Constraint (11) enforces nonnegativity.